

Sermoon S1 FAQ

About S1

1. What are the standout features of the S1 compared to previous scanners?
 - (1) Deep-hole scanning capability;
 - (2) Larger single-frame field of view: –so require fewer markers (minimum 4 markers per frame need to be recognized)
 - (3) Higher frame rate
 - (4) Innovative quad-camera system – The newly added wide-angle camera enables faster marker-free scanning of large objects and human bodies.
2. How much faster is the S1 for scanning same-sized models?

Parallel and single-line modes: 90 fps
cross-line mode: 70 fps
measurement points per second: 1.42 million (115% improvement over Raptor Pro);
infrared mode: 30 fps.
3. Is the S1 user-friendly and wireless?
 - (1) The new software offers enhanced functionality, easier marker recognition, and broader scanning range. Stronger lasers improve data capture.
 - (2) Currently wired-only but compatible with Raptor series ScanBridge.
4. Recommended applications for the S1?

The S1 excels in various scenarios, such as scanning human bodies and industrial components. It easily captures black, shiny, and metallic objects. For transparent or highly reflective surfaces, a scanning spray may be required. Its specialized deep-hole scanning mode also efficiently captures details in narrow gaps, grooves, and deep cavities.
5. Is the previous handle compatible with the S1 device?

Yes, with the Raptor series ScanBridge. Just update the firmware for compatibility.

About software

1. Failed to update directly from V4.1.1.10 (or earlier) to V4.1.1.17.

Some computers impose strict security restrictions on our software download path, causing this issue. Versions after V4.1.1.10 have updated paths to resolve it. If encountered, please manually install the newest software. Download link: <https://www.crealitycloud.com/downloads/software/creality-scan/Windows> 
2. Can scanned data be directly converted to STP?

The scanning software does not support direct export to STP format (quad-mesh data). Use third-party software (e.g., Geomagic Wrap, Geomagic Design X, or QuickSurface) for conversion. Note: Direct conversion may cause surface structure issues; reverse engineering is recommended for standardized models.

3. Does the software support dimensional inspection?

Not currently. The software provides dimensional data after scanning. Export the mesh as STL and use third-party softwares (e.g., Geomagic Control X, Polyworks, QuickSurface) for inspection.

4. Are our scanned data compatible with other reverse engineering software?

Yes. Export data as STL and import it into software like UG, QuickSurface, or Geomagic Design X.

5. Why does the software get stuck on the loading screen?

The computer configuration may be too low. Check the official Wiki <https://wiki.creality.com/en/3d-scanner/tutorials/general/performance>

About scanning technique

1. Why is calibration slow or unsuccessful?

Our calibration function is already optimized. If issues persist, ensure the setup follows the instruction diagram. You can refer to: <https://wiki.creality.com/en/3d-scanner/crealityscan/calibration>

2. Why is there blue light but no laser line during scanning?

Blue light mode requires markers. If blue light is on but no laser line appears, the scanner may not detect markers. Notes: Do not use the calibration board for scanning—its points are not markers. Store the calibration board properly as it is a precision instrument.

3. How to adjust parameters based on scanning needs?

Parameters must be tailored to the object's specifics—no fixed values. Advanced parameters may require higher computer specs.

4. When to use markers? How to apply them correctly?

Markers are mandatory in blue laser mode. In infrared mode, use markers if the object lacks geometric/textural features for tracking. Apply markers on flat or low-curvature surfaces: larger markers with wider spacing for large planes, smaller markers with tighter spacing for complex details. Avoid covering features like holes or engravings. Ensure irregular distribution. For details, see: <https://wiki.creality.com/3d-scanner/tutorials/general/marker>

5. Why are there holes or orange surfaces after meshing?

- (1) Dead-angle structures on the object may cause data loss.
- (2) If the point cloud confidence percentage is too low, low-density areas may be deleted.